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Information Technology Competencies, Organizational Agility, and Firm Performance: Enabling and Facilitating Roles

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The hypercompetitive aspects of modern business environments have drawn organizational attention toward agility as a strategic capability. Information technologies are expected to be an important competency in the development of organizational agility. This research proposes two distinct roles to understand how information technology competencies shape organizational agility and firm performance. In their enabling role, IT competencies are expected to directly enhance entrepreneurial and adaptive organizational agility. In their facilitating role, IT competencies should enhance firm performance by helping the implementation of requisite entrepreneurial and adaptive actions. Furthermore, we argue that the effects of the dual roles of IT competencies are moderated by multiple contingencies arising from environmental dynamism and other sources. We test our model and hypotheses through a latent class regression analysis on data from a sample of 109 business-to-business electronic marketplaces. The results provide support for the enabling and facilitating roles of IT competencies. Moreover, we find that these dual effects vary according to environmental dynamism. The results suggest that managers should account for (multiple) contingencies (observed and unobserved) while assessing the effects of IT competencies on organizational agility and firm performance.

Key words: organizational agility, IT competencies, latent class regression

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1. Introduction

Hypercompetition, or intensive competitive rivalry, brought about by globalization and rapid changes in consumer preferences and economic cycles, has galvanized firms toward agility and frequent competitive action as a strategic imperative. Agility refers to a firm's ability to sense opportunities for competitive action and marshal the necessary resources to seize those market opportunities (Goldman et al. 1995). Sambamurthy et al. (2003) extend this definition by proposing that agility, as a dynamic capability, reflects an ability to detect and seize competitive opportunities with speed and surprise. Overby et al. (2006, see also Lee et al. 2009) further propose that agility encompasses firms' ability to (1) manifest entrepreneurial innovation actions in the form of new products, services, or business models that create new competitive advantages or disrupt competitors'

prevailing advantages, and (2) be resilient and adaptive to threats or challenges arising from regulatory or environmental changes, changes in consumer preferences or technologies, and competitive actions.

The extant conceptual frameworks propose that firms' IT competencies enable their agility (Sambamurthy et al. 2003). These competencies include both aspects of firms' IT infrastructure and the capabilities for selecting, acquiring, configuring, and implementing IT in the organization. Sambamurthy et al. (2003) suggest that IT competencies create digital options in the form of digitized processes and knowledge repositories, which then enable organizational agility. Agility in turn should enhance firm performance, including their profitability (e.g., Lee et al. 2009), competitive position (e.g., Lee et al. 2009), and competitive actions (e.g., Chi et al. 2010). Thus Sambamurthy et al. (2003) predict

a nomological network of relationships involving IT competencies, agility, and firm performance that deserve empirical scrutiny.

Recent empirical research tests several elements of the conceptualized nomological relationships. Lu and Ramamurthy (2011) provide evidence of the links between IT competencies and organizational agility; Tallon and Pinsonneault (2011) consider the relations between agility and organizational performance, as moderated by environmental volatility. Our research aims to extend these prevailing conceptual and empirical insights in two ways. First, we conceptualize dual roles for IT competencies: a direct enabling role, in which IT competencies support (enable) organizational agility; and, a facilitative role, such that IT competencies complement agility in the efforts to launch competitive actions and seize performance gains. The direct enabling role already has been articulated, but we propose that IT competencies also influence how well firms leverage their agile capabilities to seize opportunities for innovation, adapt to changes in business conditions, or launch new competitive actions. As is the case for most dynamic capabilities, it is not sufficient to simply create agile capabilities; rather, managerial efforts must facilitate their implementation (e.g., Sirmon et al. 2007). In this regard, we propose a second role, whereby IT competencies are used in activating the firm's agile capabilities and enhancing performance. Thus, we argue that IT competencies also play a facilitative role.

Second, we propose that the enabling and facilitative roles of IT competencies are moderated by the level of environmental dynamism (Chiasson and Davidson 2005, Rai et al. 2006). Tallon and Pinsonneault (2011) demonstrate that environmental volatility moderates the link between agility and firm performance. However, consistent with a traditional approach to the study of contingencies, they examine the effects of interactions between environmental volatility and agility on firm performance. In contrast, following Sambamurthy and Zmud (1999), we argue that environmental dynamism moderates more than just one relationship; it likely affects both the enabling and facilitating roles of IT competencies in complex ways.

Traditional approaches to moderation analysis cannot account for all of the complex ways that contingency factors can interact with other constructs in a nomological model. Thus, an omitted variable bias could invalidate previous tests of significance (e.g., Grewal et al. 2013). Therefore, we model the varied, complex interactions between the contingency factors and the research constructs as sources of unobservable heterogeneity. Using finite mixture theory (e.g., Titterton et al. 1985), we estimate simultaneously multiple latent regimes of relationships between

the explanatory and dependent variables in a latent class regression model with concomitant variables (CLCRA). Different levels of our contingency (concomitant) variable, i.e., environmental dynamism, are used to simultaneously determine the odds of firms falling into one latent regime versus another as well as the effects of the enabling and facilitative roles of IT within each latent regime (e.g., Grewal et al. 2006). Our empirical analysis extracts two latent regimes with high and low levels of environmental dynamism, respectively, that allows us to explain the distinct relationships among IT competencies, agility, and firm performance within each regime.

In the next section, we define the theoretical constructs of IT competencies and organizational agility. Building on existing literature on IT competencies and organizational agility, we develop theoretical assertions about the dual roles of IT competencies as enablers and facilitators of organizational agility. We also theorize about the differential effects of IT competencies that vary with environmental dynamism. In elaborating on our research context and methodology, we detail survey data gathered from a sample of 109 business-to business (B-to-B) electronic market-making firms, as well as the latent regime (class) regression model we use to account for unobserved sources of heterogeneities. We then present our model selection and analysis results. Finally, we discuss the implications of our findings for both theory and practice.

2. Conceptual Background

Figure 1 shows our research model, linking IT competencies, agility, and firm performance, within the context of environmental dynamism as a contingency factor.

2.1. IT Competencies

As the entries in Table 1 illustrate, several conceptualizations of IT competencies are prevalent, though most studies focus on firms' IT infrastructure and IT capabilities as underlying their ability to leverage IT functionalities. In accordance with such prior literature (e.g., Aral and Weill 2007, Piccoli and Ives 2005), we conceptualize IT competencies as comprising both IT infrastructure and IT capabilities.

We define IT infrastructure as including not only physical assets, such as hardware platforms, software applications, data repositories, and other networking and object-based technologies, but also the quality and frequency of updates to all IT-related asset stocks (e.g., Bharadwaj 2000). In contrast, IT capabilities offer a measure of a firm's technical and management skills and IT practices (Aral and Weill 2007), in that they include IT knowledge and technical proficiency that reveal how best to leverage IT systems

Figure 1 Conceptual Model

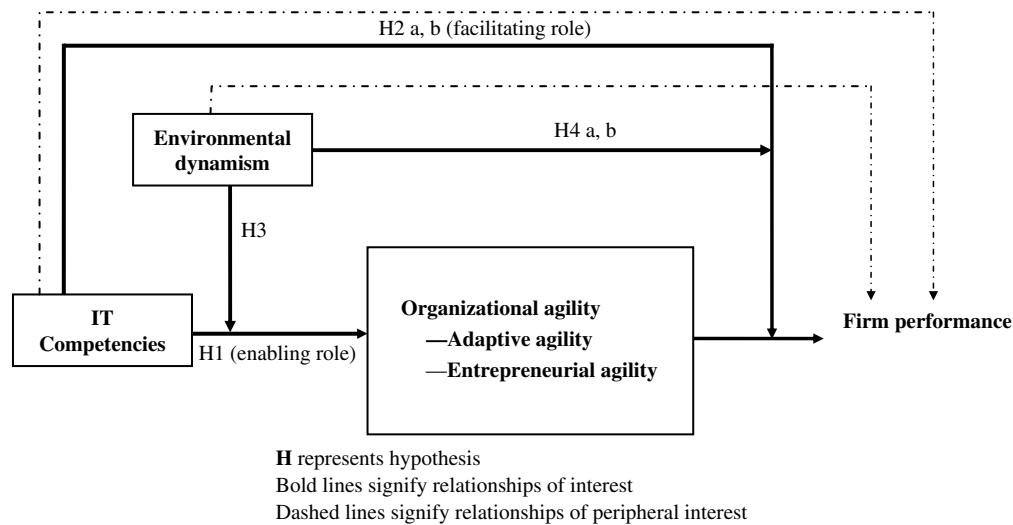


Table 1 Sample of Studies Using IT Constructs

Articles	Main construct	Dimensions of main construct
Sambamurthy et al. (2003)	IT competencies Defined as organizational base of IT resources and capabilities	—Investment scale Theoretically reflects level of investments in IT, quality of IT infrastructure. —IT capabilities Theoretically reflects IT human capital and business partnerships that enhance appropriate IT implementation.
Piccoli and Ives (2005)	Summary of constructs used to describe IT resources and capabilities	Of 39 studies summarized, 28 use IT infrastructure, 26 use IT technological skills, 23 use management skills at IT planning and implementation, 14 use information repositories, and 5 use relationship assets.
Bharadwaj et al. (1999)	IT investments Defined as creating an aggregate, uniform asset	No underlying dimensions.
Brynjolfsson and Hitt (1996)	IT investments No definition provided	—Capital Measured as self-reported spending on computer capital. —Labor stock Measured as self-reported spending on IS staff.
Aral and Weill (2007)	IT resources Defined as combinations of IT investment allocations and a mutually reinforcing system of IT competencies and practices.	—IT assets Infrastructure, transactional assets, informational assets, strategic assets. —IT capabilities IT competencies (IT skills and IT management quality) and IT practices (culture of IT use, digital transactions, and Internet architecture).

(e.g., Mata et al. 1995), as well as systems analysis, design, applications support, and project management skills that enable successful information systems implementations (Ross et al. 1996). IT capabilities also include investments in IT human skills that help identify future business needs, which are critically important for managing changes in business environments (Bharadwaj 2000, Joshi 1991).

2.2. The Multiple Forms of Organizational Agility

Lee et al. (2009) conceptualize organizational agility as a two-dimensional dynamic capability, with an entrepreneurial or offensive and an adaptive or defensive dimension. Entrepreneurial agility implies an ability to anticipate and seize market opportunities proactively and thus allows a firm to modify its positioning and strategies and organize new business approaches to gain early advantages in changing conditions. For example, buyENERGYonline.com (the United Kingdom's only surviving energy exchange for businesses) recognized the lack of regulatory help for large energy suppliers or channels for lodging complaints after the "Energy Watch" regulator disbanded in 2007. Anticipating an opportunity, buyENERGYonline.com quickly established free Internet forums that allowed energy users to post complaints in return for being able to post to the forum the names of users who might be interested in transactions (www.buyENERGYonline.com).

In contrast, with adaptive agility, the firm detects and responds to market dynamics in a defensive manner, such as protecting itself and remaining resilient, generally in an attempt to recover from disruptions in market forces rather than in response to any fundamental change in the internal structure or organization. For example, FreightX,

an existing electronic marketplace, connects shippers and carriers in the fragmented transportation industry. Although it enjoyed some success in the 1990s, the transportation industry began moving away from electronic marketplaces and toward long-term, pre-negotiated contracts, which hurt FreightX's revenues. Reacting to the changing requirements of its potential clients, FreightX began cooperating with clients to develop customized electronic trading communities that allowed shippers to manage multiple, preselected carrier relationships (Koch 2010). Such IT-based service adaptations enabled FreightX to avoid major service disruptions and grow as an electronic marketplace.

2.3. Organizational Agility and the Dual Roles of IT Competencies

IT competencies are vital for firms to sense and appropriately respond to business opportunities and challenges. However, the volume of information that needs to be processed to understand the intricacies of opportunities and challenges really demands that IT competencies help organize data. In particular, IT-based statistical tools are required to analyze, interpret, and predict how various opportunities and challenges might affect the firm and its competition. Moreover, IT-based communication tools can organize external information exchanges to support coordinated actions in the face of opportunities and challenges. By enabling real-time feedback from customers, IT-based social media tools offer filtering capabilities that organize customer opinions, leading to better anticipation of changes in market needs. Wixom and Watson (2001) emphasize that IT-based decision support systems and data warehouses help firms monitor data in real time, recognize patterns, and simulate strategic scenarios. Overall, IT competencies enable firms to sense and interpret business opportunities and challenges.

Yet IT competencies also enable firms to respond to opportunities and challenges, whether those responses are proactive or reactive in nature. Proactive responses, which lead to entrepreneurial agility, involve the ability to organize business processes to seize potential opportunities. Several aspects of IT infrastructure (e.g., IT planning capabilities) and IT skills (e.g., IT human resources) help firms foresee a wide range of IT-enabled scenarios and rapidly respond to opportunities, both of which are aspects of entrepreneurial agility. For example, vertmarkets.com proactively built a sense of community in its marketplace by regularly creating and updating downloadable libraries of white papers, electronically disseminating libraries to market participants, and organizing industry event calendars (e.g., www.vertmarkets.com, accessed March 2009). During the dot.com boom of the 1990s, preemptive

community-building moves helped avoid general discontent among market participants when vertmarkets.com began diversifying business processes across multiple industry groups (e.g., Kambil and van Heck 2002).

Reactive responses, which lead to adaptive agility, improve existing processes to adapt to the challenges posed by market changes. With IT-based communication and coordination tools, firms can increase knowledge reach and quality, such that they and their stakeholders can understand and participate in improvements to transactional processes (Overby et al. 2006). Other elements of IT infrastructure and capabilities, such as state-of-the-art IT infrastructure or technical IT skills, enable firms to defend against a wide range of disruptive IT-enabled scenarios and be resilient, both of which lead to adaptive agility. For example, in response to the heavy volume of complaints about fake transactions and less-reputable market participants, www.virtualchip.com, a B-to-B electronic trading hub for semiconductor chips, used computerized filters to qualify original equipment manufacturers, franchised distributors, and chip manufacturers using information provided by the participants themselves. This tactic helped eliminate opportunistic players and enabled the marketplace to adapt by improving its credibility. Overall, IT competencies enhance firms' response capabilities. Because IT competencies increase firms' abilities to (1) anticipate and interpret opportunities and challenges, and (2) respond to opportunities and challenges in a proactive and reactive manner, they should enable both entrepreneurial agility and adaptive agility.¹ Thus, we propose:

HYPOTHESIS 1 (H1). *Increases in IT competencies increase entrepreneurial agility and adaptive agility.*

2.3.1. Facilitating Role of IT Competencies. Traditional perspectives on strategy emphasize sustainable competitive advantages; recent perspectives propose instead that firms succeed due to their ability to execute a series of competitive actions, most of which might confer only short-term performance gains (Grimm et al. 2006). These actions include the entrepreneurial launch of new business models, products, or services; adaptive moves to respond to competitive actions by rivals; and resiliency efforts to protect positions against emerging regulatory changes. D'Aveni et al. (2010) propose that

¹ This hypothesized relationship between IT competencies and organizational agility has received some empirical scrutiny. Lu and Ramamurthy (2011) find that IT capability positively influences market capitalizing agility and operational adjustment agility; Lee et al. (2009) find that IT capability positively influences entrepreneurial and adaptive agility.

firms focus on the creation and management of temporary competitive advantages through these competitive actions. Thus, firms excel if they are able to arbitrage gains from continual competitive actions.

Sambamurthy et al. (2003) argue that as a dynamic capability, agility establishes a platform for launching competitive actions with speed, surprise, and disruption. However, the mere possession of these capabilities is not enough; they must be continually activated and exercised so that firms sense opportunities for entrepreneurship or threats that require their adaptive actions, then marshal the necessary resources and endowments to launch appropriate competitive actions. Piccoli and Ives (2005) argue that when firms operate in information-intense environments, their competitive performance depends on the interactions between IT competencies and other organizational capabilities. Prior research has demonstrated that IT competencies interact with other organizational capabilities to affect firm performance (Banker et al. 2006, Rai et al. 2006).

These perspectives underlie our conceptualization of a second, facilitating role for IT competencies. Firms can build capabilities by creating and maintaining a stock of relevant resources and skills; their ability to activate these capabilities is equally important. As the resource-based view of the firm suggests, competitive advantage accrues from the organizational ability to channel resources toward specific opportunities and challenges (Peteraf 1993). Furthermore, embedded capabilities become dynamic only if the firm can implement capabilities for sensing and responding to a changing environment (Teece 2007). Most firms build agile capabilities by creating and maintaining resources related to multiple functional competencies, including IT competencies. For example, Sanchez and Mahoney (2002) emphasize the contribution of modular new product resources that enable product customizations in competitive environments. Hitt et al. (1998) theorize that core marketing competencies, such as customer and competitor sensitivities, should enable firms to adjust to different market scenarios.

Although the possession of multiple functional competencies might help establish a capability to be agile, firms must channel their multiple competencies into implementing competitive actions in support of performance gains. As IT competencies (i.e., IT infrastructure and capabilities) increase, the business value derived from most business functions also increases (e.g., Barua et al. 1995, Pavlou and El Sawy 2006, Sethi et al. 2003). For example, brand building and product development activities can generate better payoffs if IT systems, such as customer relationship management applications, can be leveraged to

facilitate analyses of customer profile information and market demographics.

Similarly, as Sambamurthy et al. (2003) suggest, IT competencies should facilitate managerial cognition and sense making, which in turn may help firms leverage their organizational agility. For example, entrepreneurial agility enables firms to transform their assets and improve firm performance; state-of-the-art IT infrastructure and IT human resources should facilitate the speed with which such transformations take place. That is, IT competencies may facilitate the influence of entrepreneurial agility on firm performance. Moreover, adaptive agility influences firm performance by making the firm resilient and capable of absorbing environmental shocks, in that it is operations oriented, and through incremental changes, such as reduced cycle time (e.g., Sambamurthy et al. 2003). In turn, aspects of IT competencies, such as IT capabilities, should complement this operations orientation by ensuring timely information availability and processing information to enable information utilization, which facilitates firm performance. Therefore, IT competencies may enhance the influence of adaptive agility on firm performance. Building on this logic for entrepreneurial and adaptive agility, we suggest that the facilitating effects of IT competencies materialize as positive moderators of the impacts of organizational agility on firm performance. That is,

HYPOTHESIS 2 (H2). *As the degree of IT competencies increases, increasing levels of (a) entrepreneurial agility, and (b) adaptive agility lead to improved firm performance.*

2.4. The Effects of Environmental Contingencies

Consistent with Sambamurthy and Zmud (1999), we suggest that multiple contingencies affect the links among IT competencies, agility, and firm performance; these contingencies arise from the nature of the business environment and the industry where firms differ in terms of their strategic orientation, demographic characteristics (size and age), competitive position, and IT maturity. However, researchers can study only a limited number of contingencies; the complexity of the relationships among constructs increases drastically with the number of contingency factors. Although we maintain that multiple contingencies are at work, because “business firms are subject to the pulls and pressures of multiple, rather than singular, contingency forces” (Sambamurthy and Zmud (1999, p. 261), we can theoretically identify the effects of only observed contingencies. We are particularly interested in the contingency effects of environmental dynamism, which is both observed and represents variability in multiple components of the external environment, such as customers, competitors, and technology (e.g., Garg et al. 2003, Hough

and White 2003, Tallon and Pinsonneault 2011). However, we can compensate for the challenges of theoretically accounting for unobserved contingencies by modeling the numerous and complex contingencies as sources of unobservable heterogeneity in our latent regime (class) regression analysis, as we elaborate subsequently.

2.4.1. The Enabling Role of IT Competencies and the Contingency Effects of Environmental Dynamism. As environmental dynamism increases, anticipated changes often fail to materialize, and the frequency of unexpected eventualities rises (e.g., Winter 2003). The IT competencies designed to organize and analyze information for sensing and understanding environmental changes might rely too much on historic precedence and path dependencies to support effective scanning and interpretation of the environment as its dynamism increases (e.g., Pavlou and El Sawy 2010, Rindova and Kotha 2001). These IT competencies may be rigid rather than amenable to frequent reorganization and realignment with the pressing demands of the moment (Pavlou and El Sawy 2010).

Although IT competencies create digital processes based on modular and flexible IT infrastructures and provide firms with multiple options, increasing degrees of environmental dynamism lead to corresponding increases in the frequency and radicalness of environmental changes, which do not require planned digital options. Instead, they likely warrant alterations to the nature of the knowledge on which existing IT competencies are based. To maintain agility in increasingly dynamic environments, IT competencies should continuously evolve, realigning with the changing nature of knowledge and technology. Grewal and Tansuhaj (2001) emphasize this need to evolve by showing that existing, market-based, information-processing systems become ineffective as the degree of environmental dynamism increases.

The alignment of each component of IT competencies—IT infrastructure and IT capabilities—with the rest of the organization often requires a path-dependent accumulation of information repositories (e.g., Piccoli and Ives 2005) and planned architectural designs. As the degree of environmental dynamism increases, such information repositories might grow less relevant, and architectural designs might be too embedded in outdated procedures and routines to be conducive to frequent, rapid rearrangements. Thus, architectural designs sometimes hinder adjustments of business processes and restrict information usage by storing data in ways that are difficult to access or analyze (e.g., Overby et al. 2006). We suggest that as the degree of environmental dynamism increases, IT competencies may lose their ability to enable organizational agility. Because

organizational agility comprises both entrepreneurial and adaptive agility, we hypothesize,

HYPOTHESIS 3 (H3). *As the degree of environmental dynamism increases, the positive effects of IT competencies on entrepreneurial and adaptive agility decrease.*

2.4.2. The Facilitating Role of IT Competencies and the Contingency Effects of Environmental Dynamism. Although IT competencies facilitate the effective operations of business functions and processes (e.g., Barua and Mukhopadhyay 2000, Wheeler 2002), they must align with the requirements of different business functions to facilitate performance on varied dimensions, such as transactional efficiencies (e.g., operational efficiencies with the help of e-commerce platforms) and product customization (e.g., product creation modularization through product design software systems). Increasing alignment between individual business functions and IT competencies should enhance organizational performance; building on this logic, we propose that IT competencies facilitate the positive links between agility and firm performance.

Despite this prevailing alignment between IT competencies and agility and its beneficial effects on performance, we argue that enhanced environmental dynamism still could pose difficulties (Tallon and Pinsonneault 2011). For example, several B-to-B electronic marketplaces, such as Ariba.com, grew initially by attracting customers with the promise of customized IT platforms, such as electronic data interchanges, that could cater to the unique need of the customer. However, when the environment changed dramatically after September 11, 2001, and during the modern global recession, many of these B-to-B electronic exchanges that were agile in the e-commerce boom environments of the 1990s simply could not meet the fundamental requirements of existing and potential customers in a new environment (Albrecht et al. 2005). These customers demanded integrated IT systems (originally designed for individual suppliers or buyers) across multiple suppliers and buyers in electronic marketplaces, all of whom were linked to one another through a complex transactional web. Although these marketplaces were inherently agile enough to respond to the unique needs of separate customers, they could not leverage this agility when customers wanted to coordinate among themselves to meet the challenges of the looming recession. The failure to adjust its IT competencies to the agility demands of the new business environment led to the nearly immediate failure of Ariba.com after 2001.

Increasingly dynamic and nonlinear trends in the environment make business function-specific IT systems vulnerable to inflexible processing rules

and procedures, which is not conducive to large-scale realignments with different, often conflicting, embedded IT systems (Malhotra 2003). Thus, increases in the degree of environmental dynamism could increase the probability that IT competencies fail to facilitate the effects of agility on performance. We propose that as the degree of environmental dynamism increases, IT competencies might be unable to facilitate (i.e., positively moderate) the impact of organizational agility on firm performance.² Therefore, we hypothesize a three-way interaction that qualifies H₂:

HYPOTHESIS 4 (H4). *As the degree of environmental dynamism increases, the positive moderating influence of IT competencies on the impacts of (a) entrepreneurial agility, and (b) adaptive agility on firm performance decreases.*

3. Research Method

To test our conceptualization, we use the context of B-to-B electronic marketplaces, in which IT infrastructure and IT capabilities effectively facilitate strategic and tactical operations (e.g., Kaplan and Sawhney 2000, Smith et al. 2000). Because these electronic intermediaries are recent phenomena, affected by the vagaries related to Internet technologies and electronic commerce (e.g., Garicano and Kaplan 2001), understanding the complex interactions between IT competencies and organizational agility in this context should have substantive implications. We create a sampling frame from a publicly available mailing list developed by the online magazine *Net Market Makers*. This list comprises various B-to-B electronic commerce players, including independent electronic marketplaces, infrastructure providers, and industry-sponsored markets in the Jupiter Media Matrix. We also attempted to confirm the valid existence of these B-to-B electronic commerce firms by visiting their websites, then gathered a snowball sample by moving beyond the initial list. Our final sampling frame thus comprises 428 B-to-B electronic marketplaces.

We contacted the B-to-B electronic marketplaces by telephone to obtain the name and designation of a top manager involved in strategic decision making. Each potential respondent (most of whom were chief executive officers, founders, or vice presidents) received a data collection packet that included a cover letter, questionnaire, \$1 as an incentive, and a return envelope. Two follow-up contacts were attempted with

Table 2 Descriptive Profile of B-to-B Electronic Marketplaces in the Sample

Number of B-to-B electronic marketplaces in our sample	109
Number of B-to-B electronic marketplaces surviving as of January 2011	92
Number of B-to-B electronic marketplaces surviving as of January 2011 (the preceding row includes marketplaces that morphed)	75
Average age of B-to-B electronic marketplaces in the sample at the time of data collection	5.98 years
Average employee size of market makers in the sample at the time of data collection	858.7
Number of biased B-to-B electronic marketplaces at the time of data collection	75
Number of vertical B-to-B electronic marketplaces at the time of data collection	85
Number of workflow marketplaces at the time of data collection	5
Number of lead generators at the time of data collection	13
Number of auctions at the time of data collection	7
Number of reverse auctions at the time of data collection	9
Number of catalog aggregators at the time of data collection	16
Number of exchanges at the time of data collection	51
Number of other types of B-to-B electronic marketplaces at the time of data collection	8

Notes. (a) Electronic marketplaces are categorized as (1) catalog aggregators—when the market maker collects the offerings from multiple vendors and sells them to buyers at relatively static prices, (2) auctions—when multiple buyers bid for a single supplier's offerings, (3) reverse auctions—when multiple sellers bid competitively to fulfill the needs posted by a buyer, (4) exchanges—when multiple buyers and sellers interact and trade through some dynamic pricing mechanism similar to what is used in stock exchanges, (5) lead generators—when the market maker generates leads for sellers and earns revenue from advertisements, fees and commissions for finding qualified sales leads, and (6) workflow market place—when the market maker charges fees for providing collaboration services for complex multiparty projects such as construction jobs. Other types include firms that provide multiple services such as combinations of catalog aggregator and exchange, etc.

(b) An electronic marketplace (1) is biased if the market maker (i.e., the owner and administrator of the marketplace) participates in the market as a buyer and/or a seller, and (2) is vertical when it is industry-specific but horizontal if it is business-process-specific and encompasses multiple industries.

nonrespondents. We received complete and usable surveys from 109 firms. A comparison of early with late respondents in terms of sales indicated no differences ($t = 1.145, p > 0.20$), reducing concerns of non-response bias (e.g., Armstrong and Overton 1977). We provide a summary description of the sample in Table 2.

3.1. Measures

In Appendix A we present our measures. The dependent construct, firm performance, is a subjective measure of the extent to which the sales, return on investments, profits, growth, and market share goals of the electronic marketplace have been achieved. Subjective measures of performance are widely used and show high convergent validity with objective accounting measures (e.g., Sarkar et al. 2001, Worren et al. 2002). To measure entrepreneurial agility, we

² The only other empirical study of the performance consequences of organizational agility is research by Tallon and Pinsonneault (2011), who show that agility positively influences firm performance and that the magnitude of this positive effect increases as environmental volatility increases.

relied on a definition adapted from Overby et al. (2006): entrepreneurial agility is the ability to anticipate and respond to market dynamics in a proactive manner. Therefore, the measure of this construct pertains to the degree to which the firm proactively capitalizes on impending market change (i.e., anticipate market change and implement strategic actions to take advantage of it). To ensure the measurement items had sufficient face validity, we showed our list of survey items to two professionals with more than five years of experience managing e-commerce websites. After this consultation, we finalized four items as measures of entrepreneurial agility: (1) building capabilities to foresee a wide range of scenarios (i.e., being able to anticipate change), (2) rapidly taking advantage of opportunities (i.e., being able to capitalize on opportunities as they occur), (3) modifying strategic assets (i.e., being able to implement organizational change, akin to being able to respond to opportunities), and (4) easily modifying positioning strategies (i.e., being able to respond to opportunities by making strategic modifications). We assessed the psychometric properties of this construct using exploratory factor analysis with Varimax rotation. The factor loadings were all greater than 0.5, and the items showed low cross-loadings (less than 0.25) with other factors. We denote the convergent and discriminant validity of the construct in our subsequent discussion of the confirmatory factor analysis.

Similar to entrepreneurial agility, our measure of adaptive agility reflected our definition of adaptive agility, as adapted from Overby et al. (2006): the ability of a firm to detect and respond to market dynamics in a defensive manner. Therefore, adaptive agility should measure the degree to which a firm reacts to an eventuality by buffering itself and correcting for disruptions without requiring major strategic changes (e.g., Sheffi and Rice 2005). Through consultation with two professionals, we finalized a four-item measure of adaptive agility: (1) the ability to defend against a wide range of scenarios (i.e., being able to protect or buffer the firm from various disruptive forces), (2) the ability to react to abnormal conditions and resume normal activities (i.e., being able to correct for disruptions without major changes in normal activities), (3) absorption of environmental shocks (i.e., being able to withstand environmental disruption, akin to buffering itself), and (4) resiliency (i.e., being able to withstand disruptive factors, synonymous with both buffering against disruptive factors and correcting for disruptive factors without significant strategic changes). We assessed the psychometric properties of this construct using exploratory factor analysis with Varimax rotation. The factor loadings were all greater than 0.6, with low cross-loadings (less than

0.32) across items. We discuss the convergent and discriminant validity of the construct subsequently.

Because IT competencies constitute a second-order construct derived from first-order IT infrastructure (e.g., Bharadwaj 2000) and IT capabilities (e.g., Aral and Weill 2007) constructs, we used separate sets of measures. Specifically, the measure of IT infrastructure, or the quality of IT assets, contained four items: two to measure the degree of investments in and emphasis on building state-of-the-art IT infrastructure and two to capture the emphasis on planning for and updating IT assets. We used five items to measure IT capabilities, or the understanding of and technical proficiency for leveraging IT systems (e.g., Mata et al. 1995): two items related to the extent to which a firm has strong technical skills and IT-related knowledge, two items related to the extent to which the firm makes heavy investments in IT human resources, such that its developed skills are comparable to the best in the industry, and a one-item measure of the organizational understanding of how IT applications may be used effectively.

Our measure of environmental dynamism was adapted from Achrol and Stern (1988). It described the extent of changes/variability in membership in electronic marketplaces, competitive behavior, customer behavior, and the technical knowledge base. That is, our measure of environmental dynamism encapsulated variability in multiple environmental dimensions.

Finally, we used six firm-related characteristics as controls for both the organizational agility and firm performance dependent variables. Market orientation captured the extent to which a firm understands its marketplace (i.e., customers and competitors; for a review, see Kirca et al. 2005). Marketing strategy literature presents ample evidence that market orientation is a significant indicator of performance (e.g., Jaworski and Kohli 1993) and can affect the speed of strategic responses (Barrett and Weinstein 1998, Hurley and Hult 1998). Thus, we account for the market orientation of the market maker (i.e., owner and administrator of the electronic marketplace). This measure was adapted from Deshpandé and Farley (1998). We also included the age (i.e., number of years in existence) of an electronic marketplace as a control, because organizational experience and learning often is associated with age (e.g., Agarwal and Gort 2002). Thus, age could predict the manner in which firms respond to opportunities and challenges and in turn their performance. We account for firm size too; size has often been associated with the level of resources available (e.g., Mishina et al. 2004). Firm size thus could partly predict the nature of strategic responses (e.g., Chandy and Tellis 2000 show that larger firms introduce more breakthrough innovations

in dynamic environments than smaller firms) and firm performance. We take the natural logarithm of firm size to ensure a normalized distribution. We also control for the configuration of the B-to-B electronic marketplace,³ because the choice of a specific configuration can determine the kind of strategic responses available during any eventuality and therefore may also determine firm performance. Whether the B-to-B electronic marketplace is industry-specific (vertical) or business-process-specific (horizontal) might determine the range of strategic responses available during a contingency, as might the bias of a marketplace, namely, whether the market maker participates in the market as a buyer and/or seller, in that this bias could determine how well the market maker senses changes and responds (e.g., Malone et al. 1989).

3.2. Measure Validation, Common Method Bias, and Measurement Error

Because IT competencies represent a second-order latent construct, we conducted a second-order confirmatory factor analysis with the first-order constructs of IT infrastructure and IT capabilities; the model fit is acceptable ($\chi^2_{44} = 95.92$, root mean square error of approximation [RMSEA] = 0.05, confirmatory fit index [CFI] = 0.97). To ensure an adequate sample size relative to the number of estimated parameters, we used two confirmatory factor analyses to assess the convergent and discriminant validity of all our constructs. The first confirmatory factor analysis features only the dependent variables: entrepreneurial agility, adaptive agility, and firm performance (see Figure 1). The analysis reveals acceptable fit ($\chi^2_{62} = 11.46$, RMSEA = 0.04, CFI = 0.92). All items load substantively (> 0.5) and statistically significantly on their hypothesized factors. The reliabilities for the latent constructs, calculated using composite scale reliabilities (Bagozzi and Yi 1988): 0.85 for adaptive agility, 0.72 for entrepreneurial agility, and 0.93 for firm performance. Discriminant validity exists

if the average variance extracted (AVE) of a construct is greater than its squared correlation with any other construct (Fornell and Larcker 1981, Thatcher and Perrewe 2002). The AVE values were 0.55 for entrepreneurial agility, 0.61 for adaptive agility, and 0.73 for firm performance—all greater than their squared correlations. The second confirmatory factor analysis included the following independent variables: entrepreneurial agility (with firm performance as the dependent variable; see Figure 1), adaptive agility (with firm performance as the dependent variable; see Figure 1), environmental dynamism, IT competencies (such that the items for IT infrastructure and IT capabilities load on one factor), and market orientation. The model showed good fit ($\chi^2_{13} = 17.75$, RMSEA = 0.05, CFI = 0.92), with construct reliabilities of 0.85 for IT competencies, 0.85 for adaptive agility, 0.72 for entrepreneurial agility, 0.88 for environmental dynamism, and 0.97 for market orientation. The AVE value of the latent constructs were 0.81 for IT competencies, 0.55 for entrepreneurial agility, 0.61 for adaptive agility, 0.72 for environmental dynamism, and 0.88 for market orientation, all of which were greater than the squared correlations between any two factors.

To alleviate common method concerns, we applied Harmon's single-factor test, using exploratory factor analysis (Podsakoff and Organ 1986). We estimated a model of all items for the seven latent constructs (i.e., entrepreneurial agility, adaptive agility, environmental dynamism, firm performance, market orientation, IT infrastructure, and IT capabilities). The scree plot that explained 78.1% of the variance implied at least six latent factors. The resulting unrotated component matrix showed that the measurement items did not load on a single construct. The rotated factor-loading matrix indicated that the items for the different constructs loaded on different latent factors (IT infrastructure and IT capabilities load on the same factor). We also examined common method variance (CMV) using a marker variable technique. Lindell and Whitney (2001) argued that the second lowest observed positive correlation in the correlation matrix offered a reasonable estimate of CMV. If the observed correlations in the correlation matrix, when adjusted for CMV, do not differ significantly from the observed correlations, CMV generally is not substantial. In our findings, the second lowest positive correlation between any two constructs was 0.02. We adjusted all observed correlations by the method specified by Lindell and Whitney (2001) and did not find any statistically significant difference among observed and adjusted correlations ($t = 0.51, p > 0.90$).

³ The configuration of B-to-B electronic marketplaces refers to the different ways buyers and sellers transact over the exchange and the nature of the pricing mechanism, which could be static or dynamic. Accordingly, electronic marketplaces are categorized as (1) catalog aggregators, such that the market maker collects the offerings from multiple vendors and sells them to buyers at relatively static prices; (2) auctions, in which multiple buyers bid for a single supplier's offerings; (3) reverse auctions, in which multiple sellers bid competitively to fulfill needs posted by a buyer; (4) exchanges, where multiple buyers and sellers interact and trade through some dynamic pricing mechanism similar to that used in stock exchanges; (5) lead generators, for which the market maker generates leads for sellers and earns revenue from advertisements, fees, and commissions for finding qualified sales leads; and (6) workflow marketplace, in which the market maker charges fees for providing collaboration services for complex multiparty projects, such as construction jobs.

3.3. Concomitant Latent Regime (Class)

Regression Analysis

Standard (aggregate) data-analytic approaches, such as regression analysis, allow for a single set of effects of explanatory variables on dependent variables (i.e., the same regression coefficients apply to all observations in a sample). We refer to these models as single-regime models or aggregate models because there is only one regression coefficient for each explanatory variable. Single-regime models depict heterogeneity in the level of the explanatory variables and thus suggest how a change in their levels might influence the level of the dependent variable. In addition to heterogeneity stemming from the level of the explanatory variable, heterogeneity often appears in the effect (i.e., regression coefficient) of the explanatory variable on the dependent variable. Thus, for example, each explanatory variable can have two or more regression coefficients, each of which might apply to subsets of observations in a sample (e.g., Grewal et al. 2008). Because each set of regression coefficients (where a set captures the effect of all explanatory variables on the dependent variable) is a regime, we refer to these models as multi-regime models; furthermore, because we cannot identify the regimes a priori from the data, they are latent regime models, and we can model the variables that determine to which regime each observation (i.e., a firm in our case) belongs. Thus, these models entail concomitant latent class regression analyses (CLCRA; Dayton and MacReady 1988)

As a disaggregated statistical method,⁴ CLCRA reveals the number of regimes (i.e., one or more) and firm memberships in regimes (e.g., Dayton and MacReady 1988). The technique uses finite mixture distribution theory to estimate simultaneously multiple regimes of relationships between explanatory and dependent variables, as well as the coefficients for profiling variables (i.e., contingency variables such as environmental dynamism in our context), which, along with the unobserved variables, allocate firms into the latent regimes (Dayton and MacReady 1988). Consistent with emerging research that uses CLCRA to test hypotheses (e.g., Grewal et al. 2006), we use such analyses to verify whether a single-regime aggregate model is appropriate; if not, we further use CLCRA to identify (1) the appropriate number of regimes that indicate distinct relationships between independent and dependent constructs, and (2) which firms can be assigned to each regime. In other words, we do not define the regime to which a

firm belongs; the CLCRA instead allocates firms into a regime using an integrative algorithm that maximizes log-likelihood values. The optimal solution is one in which the number of regimes, regime membership for firms, and the regression coefficient for each regime minimize the information criteria (i.e., the log-likelihood value is maximized after correcting for the number of estimated parameters).

For our hypotheses testing, we note two relevant issues if the optimal model has multiple regimes (i.e., the aggregate regression model is not optimal). First, in H1 and H2, we specify only one relationship pattern among variables, but if multiple regimes exist, there also must be multiple patterns of relationships among variables, which raises questions about the appropriate hypothesis test. We argue that H1 and H2 receive support if at least one regime displays the hypothesized pattern (in line with Grewal et al. 2013, Srinivasan 2006). The presence of other regimes then would validate our contention that firms are heterogeneous and cannot be expected to behave uniformly. Second, we test H3 and H4, in which we predict contingency effects of environmental dynamism, by assigning environmental dynamism as a (concomitant) profiling variable. The levels of profiling variables likely can partly predict the allocation of firms to different regimes, relative to a base regime. For example, if we treat Regime 1 as the base regime, and environmental dynamism is a positive and statistically significant predictor of whether a firm belongs to Regime 2 relative to Regime 1, the pattern of variable relationships in Regime 1 (i.e., as hypothesized in H1 and H2) holds only at lower degrees of environmental dynamism. A different pattern of relationships in Regime 2 holds at higher degrees of environmental dynamism. We can test for statistically significant differences in the means of environment dynamism across regimes. To calculate the means, we average the values of environmental dynamism across all firms assigned to a regime. Overall, the presence of multiple regimes should not pose difficulties for hypotheses testing. For technical details on CLCRA, see Appendix B.

3.4. Measurement Invariance: Latent Class Confirmatory Factor Analysis

Before we can test whether the structural parameters (i.e., influence of explanatory variables on the dependent variable) vary across latent regimes, we must establish measurement invariance, in the sense that the measures (e.g., factor loadings, residual variances) do not vary across the latent regimes (Meredith 1993). Without measurement invariance, we cannot be sure if the latent regimes reflect structural parameter heterogeneity or their measurement invariance. Therefore, we relied on latent class confirmatory factor analysis, in which the factor loadings and error

⁴ In latent class regression analysis (e.g., Grewal et al. 2006), an aggregate method implies a single-regime regression (i.e., estimating only one relationship between independent and dependent variables), whereas disaggregated methods allow for the possibility of more than one set of relationships between the independent and dependent variables.

variances could vary across latent classes. Such an analysis is similar to CLCRA, though in CLCRA we would use a regression model, and here we have a factor analysis model. The details for the latent class confirmatory factor analysis appear in Appendix C.

4. Results

Our data analysis followed the two-step procedure espoused by Anderson and Gerbing (1988). First, we ran the measurement model to determine the appropriate number of measurement regimes and the psychometric properties of all measurement items (see §3.5). The modified Akaike information criterion (MAIC) values for the latent class confirmatory factor analysis models for all dependent variables (i.e., firm performance, entrepreneurial agility, and adaptive agility) and all independent variables (i.e., entrepreneurial agility, adaptive agility, IT competencies, environmental dynamism, and market orientation) show that a single-regime solution is optimal in all cases. We thus establish measurement invariance, in that each measurement model parameter takes only one unique (not multiple) value. Next, to account for measurement error, we compute factor scores and arrive at the aggregate measures for the latent constructs (e.g., Lastovicka and Thamodaran 1991). Following recommendations from Skrondal and Laake (2001) we use blockwise factor scores—specifically, regression factor scores for independent latent constructs and Bartlett factor scores for the dependent latent construct—then apply them in the CLCRA estimation.⁵

4.1. CLCRA Model Selection and Diagnostics

As Figure 1 and our hypotheses indicated, we considered three regression equations. The first used entrepreneurial agility as the dependent variable and IT competencies and the control variables as independent variables. The second equation featured adaptive

agility as the dependent variable and IT competencies and the control variables as the independent variables. These two equations refer to the enabling role of IT competencies. The third equation included firm performance as the dependent variable and entrepreneurial agility, adaptive agility, IT competencies, their interactions, and the control variables as the independent variables. This equation thus models the facilitating role of IT competencies. To estimate the contingency effects of environmental dynamism, we assigned environmental dynamism as a profiling variable for the CLCRA method. We estimated all three regression equations simultaneously with the CLCRA method.

With the expectation-maximization (EM) algorithm, we maximized the log-likelihood function (Dempster and Laird 1977). A potential drawback of this approach is that it is difficult to understand whether the EM algorithm converges to a local instead of a global maximum, that is, if the posterior probabilities are independent of starting values. To ensure that we obtained the optimal solution for each regime, we used the recommended 100 randomly selected starting values (e.g., Muthén 2001, Muthén and Shedden 1999), 20 iterations in the initial stage (e.g., Muthén and Muthén 2007), and as many as 2,000 total EM iterations.

The MAIC values suggested an optimal two-regime solution ($MAIC_{1 \text{ regime}} = 1,401.82$, $MAIC_{2 \text{ regime}} = 1,344.98$, $MAIC_{3 \text{ regime}} = 1,353.54$). The entropy of separation of the two-regime solution of 0.85 indicated a good separation of latent regimes (Celeux and Soromenho 1996). We also considered model identifiability, or whether there was one-to-one correspondence between the probability distribution of the data and the values of the model parameters. If a model were not identified, (1) multiple regimes may have similar estimates of the parameters of interest, which confounds the interpretation of results, and/or (2) there may be at least one regime to which extremely few firms (close to 0) can be assigned on the basis of posterior probabilities (Leisch 2004). A comparison of parameter estimates (see Table 4) between the two regimes clearly indicated that our two-regime model was identifiable, because the two regimes showed distinct relationships for the variables of interest, and each regime included a substantial proportion of firms in the sample (Regime 1 had 67% of firms, or 73, and Regime 2 had the other 33%, or 36 firms). Moreover, by establishing measurement invariance, we ruled out the possibility that the distinct relationships between dependent and independent variables in the two latent regimes could be confounded by multiple regimes of the measurement model parameters. We present the descriptive statistics in Table 3.

⁵ For the explanatory latent variables, the regression factor scores can be calculated as (Thomson 1934) $A_{\xi} = [\Phi \Lambda_x' (\Lambda_x \Phi \Lambda_x')^{-1} X']'$, where ξ represents the explanatory latent variables; A_{ξ} is the $(n \times k)$ matrix of factor scores for the explanatory latent variables, such that n is the sample size and k is the number of explanatory latent variables; Φ ($k \times k$) is the covariance matrix of ξ ; Λ_x is the matrix of factor loadings ($q \times k$), such that q is the number of items used to measure the k explanatory latent variables; and X is the $(n \times q)$ data matrix of items for the explanatory variables. For the dependent variables, the Bartlett factor scores can be calculated as (Bartlett 1937) $A_{\eta} = (\Lambda_y' \Theta_e^{-1} \Lambda_y)^{-1} \Lambda_y' \Theta_e^{-1} Y'$, where η represents the dependent latent variables; A_{η} is the $(n \times j)$ matrix of factor scores for the dependent latent variables, such that n is the sample size and j is the number of dependent latent variables; Λ_y is the matrix of factor loadings ($p \times j$), such that p is the number of items used to measure the j dependent latent variables; Θ_e is the $(p \times p)$ covariance matrix of errors of measurement for items used to measure the dependent latent variables; and Y is the $(n \times p)$ data matrix for the items of the dependent variables.

Table 3 Descriptive Statistics

	PER	ITC	AA	EA	MO	BIAS	HOR	WM	LG	AUC	RAU	CATG	EXC	MA	ED	MS
PER																
ITC	0.22															
AA	0.25	0.17														
EA	0.26	0.32	0.44													
MO	0.28	0.18	0.33	0.32												
BIAS	0.01	−0.03	0.02	0.08	−0.01											
HOR	0.33	0.10	0.12	0.27	0.10	0.03										
WM	−0.03	0.10	−0.12	−0.16	−0.01	0.03	−0.11									
LG	−0.03	−0.22	−0.01	−0.08	−0.01	0.10	−0.05	−0.08								
AUC	−0.02	−0.22	0.10	0.13	0.01	−0.08	0.13	−0.05	−0.09							
RAU	0.15	−0.01	−0.19	0.08	−0.04	0.11	0.01	−0.06	−0.11	−0.07						
CATG	−0.15	0.03	−0.01	−0.02	−0.06	−0.25	0.17	−0.11	−0.18	−0.13	−0.15					
EXC	0.12	0.17	0.06	−0.01	0.04	0.10	−0.12	−0.22	−0.36	−0.25	−0.30	−0.50				
MA	0.09	−0.10	0.01	−0.08	−0.06	0.07	0.25	0.11	−0.06	0.21	−0.10	0.05	−0.07			
ED	−0.20	0.12	0.11	0.24	0.01	0.10	0.20	−0.10	0.23	−0.04	0.06	0.03	−0.09	−0.09		
MS	−0.12	0.09	0.12	0.01	−0.07	0.03	−0.22	−0.02	−0.05	−0.02	−0.03	−0.05	0.11	−0.02	−0.08	
Mean	4.26	5.45	4.56	4.62	4.98	69%	22%	4.5%	12%	6.4%	8%	14.6%	46.7%	5.98	4.01	858.7
Std. dev	1.35	1.03	1.04	1.01	1.12									10.83	1.07	125.73

Notes. PER = performance, ITC = information technology competencies, AA = adaptive agility, EA = entrepreneurial agility, MO = market orientation, BIAS = biased electronic market, HOR = horizontal electronic market, WM = workforce marketplace, LG = lead generator, AUC = auction, RAU = reverse auction, CATG = catalog aggregator, EXC = exchange, MA = market age, ED = environmental dynamism, and MS = size of the market maker.

4.2. Latent Regime Regression Results

We mean-centered the explanatory variables for the analysis. All variance inflation factors were lower than 10, and the condition numbers were lower than 15; multicollinearity thus was not an issue. To describe the results, we use a generic b to refer to coefficients estimated using CLCRA (b_r in Equation (B1) in Appendix B), with superscripted regime numbers and subscripted variable names (see Table 4). We focus on the theoretical relationships of interest.⁶

4.2.1. Tests of the Enabling Role of IT Competencies H1 and the Contingency Effect of Environmental Dynamism H3. In support of H1 in Regime 1, we found positive, statistically significant effects of IT competencies on entrepreneurial agility ($b_{ITC}^1 = 0.44, p < 0.01$) and adaptive agility ($b_{ITC}^1 = 0.54, p < 0.01$). That is, in 67% of our sample, we uncovered evidence of an enabling role of IT competencies. However, the results pertaining to Regime 2 rejected H1: IT competencies negatively influenced entrepreneurial agility ($b_{ITC}^2 = -0.86, p < 0.01$) and did not significantly affect adaptive agility ($b_{ITC}^2 = -0.26, p > 0.91$).

To determine why the relationships in Regime 2 did not match those in Regime 1, we examined the profiling coefficient for environmental dynamism. If we treat Regime 1 as the base, the coefficient (i.e., β_{pr} in

Equation (B3) in Appendix B) reveals that the influence of environmental dynamism on the posterior probability that any firm belongs to Regime 2, relative to Regime 1, was positive and statistically significant ($\beta_{ED} = 0.87, p < 0.01$). Thus, firms that encountered higher levels of environmental dynamism were more likely to exhibit the relationships in Regime 2 than were firms that encountered lower levels of environmental dynamism. Furthermore, the means of environmental dynamism in the two regimes differed significantly ($p < 0.05$), such that Regime 2 ($M = 4.21$) indicated a higher level of environmental dynamism than Regime 1 ($M = 3.48$). Overall, the enabling effects of IT competencies on both entrepreneurial and adaptive agility held for lower levels of environmental dynamism in Regime 1 but not at higher levels of environmental dynamism in Regime 2, in support for H3.

4.2.2. Tests of the Facilitating Role of IT Competencies H2 and the Contingency Effect of Environmental Dynamism H4. In Regime 1, we found positive, statistically significant moderating effects of IT competencies on the relationship between entrepreneurial agility and firm performance ($b_{ITC*EA}^1 = 0.23, p < 0.10$) but not between adaptive agility and firm performance ($b_{ITC*AA}^1 = -0.03, p > 0.62$). Therefore, in a majority of cases, we confirmed H2a but not H2b. In Regime 2, we found no support for H2a, considering the negative (whereas we hypothesized positive) moderating effects of IT competencies on the relationship between entrepreneurial agility and firm performance ($b_{ITC*EA}^2 = -1.29, p < 0.01$). We also

⁶ To ensure that the multiple types of respondents (vice-presidents, CEOs, founders) did not confound our results, we used three dummy variables to reflect the type of respondent and re-estimated our model using those dummy variables (with vice-president as the base) as additional controls. The results did not change in terms of the direction or statistical significance of the parameter estimates.

Table 4 Results of the Two-Regime Solution from Concomitant Latent Class Regression Analysis (CLCRA)

Variable name		DV = Entrepreneurial agility		DV = Adaptive agility		DV = Firm performance	
		Regime 1	Regime 2	Regime 1	Regime 2	Regime 1	Regime 2
	<i>Intercept Term</i>	−1.16** (0.61)	0.51 (0.61)	−0.50 (0.50)	0.16 (0.52)	−0.27 (0.92)	−2.07*** (0.88)
Relationships of interest for H1 and H3	<i>IT Competencies (ITC)</i>	0.44*** (0.13)	−0.86*** (0.19)	0.54*** (0.11)	−0.26 (0.25)	0.81 (0.83)	0.60 (0.78)
	<i>Entrepreneurial Agility (EA)</i>					2.24*** (0.22)	1.33*** (0.26)
Relationships of interest for H2a, H4a	<i>EA × ITC</i>					0.23* (0.13)	−1.29*** (0.35)
	<i>Adaptive Agility (AA)</i>					−0.17 (0.48)	0.24 (0.35)
Relationships of interest for H2b, H4b	<i>AA × ITC</i>					−0.03 (0.12)	0.53 (0.49)
Controls							
Marketplace configuration	<i>Catalogue Aggregators (CA)</i>	−0.33 (0.36)	−0.33 (0.36)	0.43 (0.48)	0.43 (0.48)	0.20 (0.46)	0.20 (0.46)
	<i>Auctions (AU)</i>	1.04*** (0.48)	1.04*** (0.48)	1.29*** (0.42)	1.29*** (0.42)	−1.99*** (0.82)	−1.99*** (0.82)
	<i>Reverse Auctions (RA)</i>	0.23 (0.37)	0.23 (0.37)	−0.22 (0.45)	−0.22 (0.45)	0.01 (0.01)	0.01 (0.01)
	<i>Exchanges (EXC)</i>	−0.01 (0.31)	−0.01 (0.31)	−0.42 (0.38)	−0.42 (0.38)	0.20 (0.21)	0.20 (0.21)
	<i>Lead Generators (LG)</i>	−1.92*** (0.43)	−1.92*** (0.43)	−0.36 (0.58)	−0.36 (0.58)	0.40 (0.39)	0.40 (0.39)
	<i>Workflow Marketplaces (WM)</i>	0.18 (0.49)	0.18 (0.49)	0.63*** (0.21)	0.63*** (0.21)	0.04 (0.22)	0.04 (0.22)
	<i>Market Orientation (MO)</i>	0.33*** (0.11)	0.33*** (0.11)	0.86*** (0.16)	0.86*** (0.16)	1.53*** (0.39)	1.53*** (0.39)
Other controls	<i>Horizontal Market (HM)</i>	0.49** (0.23)	0.49** (0.23)	0.12 (0.22)	0.12 (0.22)	0.24 (0.30)	0.24 (0.30)
	<i>Biased Market (BM)</i>	0.18 (0.19)	0.18 (0.19)	0.02 (0.13)	0.02 (0.13)	−0.16 (0.27)	−0.16 (0.27)
	<i>Market Age (MA)</i>	−0.02** (0.01)	−0.02** (0.01)	−0.01 (0.01)	−0.01 (0.01)	0.03*** (0.01)	0.03*** (0.01)
	<i>Market Maker Size (MS)</i>	0.15 (0.11)	0.15 (0.11)	0.26** (0.14)	0.26** (0.14)	0.01 (0.29)	0.01 (0.29)
	<i>Environmental Dynamism (ED)</i>	0.11** (0.05)	0.10** (0.05)	0.05 (0.18)	0.15 (0.37)	−0.51 (0.73)	−0.56 (0.79)
Regime-specific pseudo R-square		0.73	0.65	0.64	0.62	0.65	0.65
Concomitant profiling variables: Determination of regime membership (The multinomial logit part of the model)							
		Regime 1				Regime 2	
Intercept		...				−0.72*** (0.26)	
Environmental dynamism		...				0.87*** (0.35)	
Regime size		73 firms				36 firms	

Notes. Standard errors are reported in parentheses. Biased electronic markets were coded as 0, and unbiased electronic markets were coded as 1. Vertical electronic markets were coded as 0, and horizontal electronic markets were coded as 1. To capture the effect of market configuration, we treated others as the base and coded dummy variables for other configurations. We used two-tailed tests of statistical significance, due to our limited sample size. The most likely error when the sample size is small is Type-II error (i.e., rejecting true relationships). Therefore, our statistically significant results are quite conservative. The relationships of interest when entrepreneurial and adaptive agility are the dependent variables are coded in light grey. The relationships of interest when firm performance is the dependent variable are coded in dark grey. DV = dependent variable.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

did not find support for H2b in the statistically nonsignificant moderating effects of IT competencies on the relationship between adaptive agility and firm performance ($b_{ITC*AA}^2 = 0.53, p > 0.27$).

This lack of support for a facilitating effect of IT competencies in Regime 2, but partial support for it in Regime 1, led us to examine the contingency factors that underlie the differences in these regimes. As we have discussed, firms in Regime 1 encountered less environmental dynamism than those in Regime 2. In turn, we found facilitating effects of IT competencies in the relationship between entrepreneurial agility and firm performance only at lower levels of environmental dynamism (i.e., in Regime 1). As environmental dynamism increased, the facilitating effect disappeared (and even became negative), as in Regime 2. These findings support H4a. However, the statistically nonsignificant moderating effects of IT competencies on the relationship between adaptive agility and firm performance in both regimes led us to reject H4b.⁷

4.2.3. Comparison of Results Between Pooled Regression and CLCRA. Our primary motivation for using CLCRA was to demonstrate that, despite our theorizing a certain pattern of relationships among IT competencies, organizational agility, and firm performance, firms actually might show heterogeneous relationships due to observed (e.g., environmental dynamism) and unobserved contingences. Our results have revealed that a two-regime solution is optimal and outperforms an aggregate, single-regime solution. However, in this section we present a single-regime, pooled regression to demonstrate that the pooled regression model may not always be appropriate, because it masks nuances in the relationships among explanatory and dependent variables (i.e., multiple regimes might exist).

In Table 5, the pooled regression (i.e., aggregate single-regime model) results reject H1; there are no statistically significant enabling (main) effects of IT competencies on either entrepreneurial agility ($b_{ITC} = 0.18, p > 0.84$) or adaptive agility ($b_{ITC} = 0.10,$

$p > 0.36$). Nor did we find support for H2a and H2b, in that there are no statistically significant facilitating (moderating) effects of IT competencies on either the impact of entrepreneurial agility on firm performance ($b_{ITC*EA} = 1.05, p > 0.17$) or the impact of adaptive agility on firm performance ($b_{ITC*AA} = -0.06, p > 0.29$). In contrast, in the CLCRA estimation, we uncovered support for H1 and H2a in Regime 1 but not in Regime 2. Because firms clearly belong to two different regimes, marked by contrasting relationships among IT competencies, organizational agility, and firm performance, aggregating all firms in the sample for a pooled regression prevents any particular effect from dominating, and we are left with statistically nonsignificant effects. With the help of CLCRA though, we could reveal the nuances in the data.

We treat environmental dynamism as a profiling variable in the CLCRA. Therefore, our estimation model contained neither two-way nor three-way interaction terms with environmental dynamism in the pooled regression results in Table 5. We thus cannot compare the pooled regression and CLCRA with regard to the results related to H3 and H4 (i.e., contingency effects of environmental dynamism).

5. Discussion

The recipe for organizational success in modern business environments increasingly calls for agility as an important ingredient. Yet, we lack a comprehensive understanding of how organizations build and leverage agility for superior performance. Although some scholars suggest that IT competencies can help build agile capabilities (e.g., Piccoli and Ives 2005, Rai et al. 2006, Sambamurthy et al. 2003), there has been limited empirical research on the role of IT competencies. We extend knowledge about the links between IT competencies, organizational agility, and firm performance through our conceptualization and empirical research. First, we demonstrate empirically that IT competencies enable (i.e., build) organizational agility, whether the agile capabilities are entrepreneurial or adaptive in nature. Second, we conceptualize and empirically substantiate how IT competencies facilitate the positive effects of entrepreneurial agility capabilities on firm performance. Third, we show that the associations among IT competencies, organizational agility, and firm performance are not uniformly applicable to all firms. With our CLCRA methodology,⁸ we can

⁷ It is possible that IT competencies and the agility constructs are endogenous, because higher performing firms might seek greater agility and IT competencies. We thank an anonymous reviewer for bringing this issue to our attention. To correct for such endogeneity, we used the control function approach espoused by Garen (1984). We separately regressed entrepreneurial agility, adaptive agility, and IT competencies on several factors (e.g., firm size, firm age, market orientation, marketplace configuration, bias of the marketplace, whether the marketplace is vertical or horizontal). We used both the residuals of these equations and the multiplicative products of each residual with each endogenous variable as controls in the main equation, with firm performance as the dependent variable. Although we found statistically significant effects of two controls, the main results still hold in both direction and statistical significance.

⁸ We use the two-step approach espoused by Anderson and Gerbing (1988), largely because of our small sample size relative to the number of parameters we estimated (though we correct for measurement error by using factor scores). If sample size constraints did not exist, researchers could use latent class structural equation models of multiple structural regimes (see Muthén 1989). We thank an anonymous reviewer for this suggestion.

Table 5 Results from Pooled Regression

Variable name		DV = Entrepreneurial agility	DV = Adaptive agility	DV = Firm performance
Relationships of interest for H1	<i>Intercept Term</i>	−1.13 (1.23)	−1.05** (0.65)	−1.69* (0.94)
	<i>IT Competencies (ITC)</i>	0.18 (0.13)	0.10 (0.11)	0.42 (0.38)
	<i>Entrepreneurial Agility (EA)</i>			1.05 (1.44)
Relationship of interest for H2a	<i>EA × ITC</i>			−0.06 (0.18)
	<i>Adaptive Agility (AA)</i>			−0.75 (1.02)
Relationship of interest for H2b	<i>AA × ITC</i>			0.19 (0.18)
Controls				
Marketplace configuration	<i>Catalogue Aggregators (CA)</i>	0.15 (0.27)	0.05 (0.23)	−0.06 (0.31)
	<i>Auctions (AU)</i>	0.30 (0.29)	0.20 (0.29)	−0.21 (0.39)
	<i>Reverse Auctions (RA)</i>	0.28 (0.22)	−0.78*** (0.36)	0.88*** (0.44)
	<i>Exchanges</i>	0.30 (1.01)	0.10 (0.09)	0.30 (1.27)
	<i>Lead Generators (LG)</i>	0.14 (0.38)	0.22 (0.35)	0.23 (0.36)
	<i>Workflow Marketplaces (WM)</i>	0.09 (1.01)	0.18 (0.69)	0.05 (1.03)
	<i>Market Orientation (MO)</i>	0.28*** (0.11)	0.29*** (0.10)	0.17 (0.55)
Other controls	<i>Horizontal Market (HM)</i>	0.45** (0.24)	0.24 (0.23)	0.86*** (0.32)
	<i>Biased Market (BM)</i>	0.09 (0.22)	0.03 (0.16)	0.19 (0.20)
	<i>Market Age (MA)</i>	−0.01 (0.01)	0.01 (0.01)	0.02 (0.01)
	<i>Market Maker Size (MS)</i>	0.16 (0.10)	0.27** (0.12)	0.01 (0.29)
	<i>Environmental Dynamism (ED)</i>	0.33** (0.15)	0.19 (0.17)	0.06 (0.37)
Regime-specific pseudo <i>R</i> -square		0.33	0.33	0.33

Notes. Standard errors are reported in parentheses. Biased electronic markets were coded as 0, and unbiased electronic markets were coded as 1. Vertical electronic markets were coded as 0, and horizontal electronic markets were coded as 1. To capture the effect of market configuration, we treated workforce marketplaces as the base and coded dummy variables for other configurations. We used two-tailed tests of statistical significance. The relationships of interest when entrepreneurial and adaptive agility are the dependent variables are coded light grey. The relationships of interest when firm performance is the dependent variable are coded dark grey. DV = dependent variable.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

account for multiple unobserved contingencies⁹ and show that the enabling effects of IT competencies for both entrepreneurial and adaptive agility decrease

⁹ The two regimes differ in terms of dynamism; however, we might have found a two- or more regime solution to be ideal. In this case, it is plausible that we might have found the two regimes did not differ in terms of environmental dynamism (even in our two-regime solution, the regimes could be similar in their environmental dynamism). Thus, because we do not force regimes to be high or low on any variable and let the data tell the story, we model for unobserved contingencies.

as environmental dynamism increases. We also show that the facilitating effects of IT competencies for entrepreneurial agility diminish with greater environmental dynamism.

Because B-to-B electronic marketplaces are unique, technology-driven contexts, it may not be obvious to generalize our findings to brick-and-mortar contexts. The similarities and differences between electronic and brick-and-mortar contexts and their effects on the relationships among IT competencies, organizational agility, and firm performance is a promising area for

further research. Even by themselves though, B-to-B electronic marketplaces represent billions of dollars of commerce; the United States Census Bureau (2010) considers B-to-B e-commerce a \$3.4 trillion market. More than 92% of all e-commerce sales are B-to-B, and approximately 39% of manufacturer's shipments are e-commerce driven. The tremendous impact of B-to-B e-commerce suggests that our estimates of relationships among IT competencies, organizational agility, and firm performance in the context of B-to-B electronic marketplaces are valuable.¹⁰

Our findings offer both theoretical and methodological contributions to IS literature. In particular, the results pertaining to entrepreneurial agility, as it is enabled and facilitated by IT competencies in most firms in the sample (Regime 1), emphasize a broader role of IT competencies than has been theorized thus far. Several scholars suggest that IT competencies might indirectly influence firm performance by enabling organizational agility (e.g., Sambamurthy et al. 2003). In line with the possibility that IT competencies might indirectly influence firm performance, we theorize facilitating (not just enabling) roles of IT competencies for organizational agility. Unlike other business functions, such as marketing and operations, that together with IT competencies enable a base of agile capabilities,¹¹ IT competencies also are unique, in that they not only create dynamic capabilities directly (e.g., entrepreneurial agility) but also help activate those capabilities for competitive actions. This latter role is possible because most business functions that contribute to a firm's agility require some degree of IT-based support for effective implementation. Furthermore, IT competencies are so intricately embedded in organizations that most capabilities necessarily are facilitated by IT competencies (e.g., Kohli and Grover 2008). However, we also note two discrepant results: We do not find evidence of direct effects of IT competencies on firm performance, nor

do we find any facilitating effects of IT competencies on adaptive agility.

Both these nonfindings likely are artifacts of our unique context. That is, we examine nonsignificant direct effects of IT competencies on firm performance. IT is inherent to the creation of B-to-B electronic marketplaces, as well as to driving transactions within an intricate web of buyers and sellers. That is, IT is necessary for the very existence of these marketplaces. However, simply possessing a base of IT competencies may not be enough for superior performance. Rather, superior performance demands that B-to-B electronic marketplaces use their base of IT competencies to enable and facilitate dynamic capabilities, such as entrepreneurial agility.

Our second nonfinding, pertaining to the facilitation effect of IT competencies for adaptive agility, leads us to inspect our context further. These B-to-B electronic marketplaces represent contexts that are not really relationship based (i.e., there are no face-to-face interactions), and they undergo constant flux because market participants can join and leave the marketplace voluntarily at any time. Efficiency-driven firms tend to be opportunistic about availing themselves of the benefits of electronic marketplaces (Kambil and van Heck 2002). Electronic marketplaces primarily consist of corporate participants that are willing to trade in big-ticket items, such as capital goods (Malone et al. 1989); the critical performance drivers of these marketplaces are customer service and a sense of trust among the market participants (e.g., Pavlou and Gefen 2004). To foster trust, market makers (i.e., owners and administrators of marketplaces) must constantly monitor and anticipate the intentions of existing participants, as well as identify and pursue opportunistic actions from the start. For customer service, market makers should proactively upgrade transactional efficiencies for existing and potential participants, as well as restructure transactional rules and procedures in accordance with structural changes in the industries served. Overall, the requirements for performance evolve continuously, in line with market dynamics. Entrepreneurship agility (i.e., anticipate market changes and take advantage of opportunities and challenges) is much better suited than adaptive agility (i.e., recover from market changes without major adjustments to internal processes and structures) to effective performance in these contexts. Thus, it is not surprising that we do not see main effects of adaptive agility on firm performance or any facilitating effects of IT competencies for adaptive agility. Further research might investigate the differential utility of adaptive versus entrepreneurial agility in varied contexts, as well as explore whether relationships of IT competencies and organizational agility vary with the type of agility.

¹⁰ Although our findings reflect the early days of B-to-B electronic marketplaces, they also are of relevance today, for two reasons. First, our findings can apply to initial, emerging business models, and advances in IT are likely to facilitate new business models. Second, we base our theorizing on extant literature, devoid of context; thus, we could have tested our hypotheses in a context other than B-to-B electronic marketplaces. Our findings can be seen as general, though clearly replications are needed to affirm such generalizations.

¹¹ We do not specifically test for the direct impact of multiple business functions on organizational agility constructs. Our claim that IT competencies are one of many functional components that enable agility is based on (1) literature from diverse domains, such as marketing and innovation (Hitt et al. 1998, Sanchez and Mahoney 2003), that describes flexibilities derived from business functions, and (2) the statistically significant positive main effects of market orientation on both adaptive and entrepreneurial agility in both regimes of our CLCRA estimation (see Table 4).

One of the extensions of our research would be the inclusion of more aspects of IT competencies in examining their enabling and facilitating roles. While IT infrastructure and IT skills represent the most fundamental elements of most firms' IT competencies, other competencies are beginning to become salient. In particular, as firms outsource their IT assets and services and architect multisourcing arrangements, their ability to manage sourcing is a critical competency (Bapna et al. 2010). Therefore, future research should consider the role of sourcing capabilities as another salient IT competency.

Our findings also highlight our methodological contributions to IS literature. We propose a CLCRA estimation that confirms that associations among IT competencies, organizational agility, and firm performance are complex, because they are rooted in multiple contingencies (e.g., Sambamurthy and Zmud 1999), most of which we treat as unobserved. Typically, when multiple contingencies exist and the research uses aggregate data-analytic approaches such as regression analysis or standard structural equation models, aggregation biases also arise, because the theoretical effects of interest differ among firms that face different levels of a contingency, such as environmental dynamism. In turn, at an aggregate level, the different effects may cancel out, leading to statistically nonsignificant effects. The power of disaggregate data-analytic techniques, such as CLCRA, is evident from the results that show two regimes and substantial differences in the results between regimes. Profiling variables in CLCRA provides insights about sources of heterogeneity between latent regimes. For example, relative to Regime 1, the degree of environmental dynamism encountered by B-to-B electronic markets in Regime 2 is significantly greater. Even if information about profiling variables were unavailable, we could account for unobserved heterogeneity by including only an intercept term in the concomitant variable part of the model (i.e., Equation (3)). Researchers thus might use CLCRA models at the end of their aggregate analysis, to verify the robustness of their results against multiple unobserved contingencies.

5.1. Managerial Implications

We spoke directly to managers of market making firms that own and administer B-to-B electronic marketplaces. Although these electronic marketplaces are built on IT, the role of IT is not limited to the infrastructural base of these marketplaces. In an environment in which face-to-face interactions among market participants are not possible, managers should build their IT competencies, with the knowledge that their most important tool for navigating the pressures and demands of the marketplace is their ability to make effective use of IT competencies. The effective

use of IT competencies requires an awareness of two things. First, IT enables agility, so it can be used to respond directly to a contingency. For example, to prevent opportunistic behavior by market participants, tracking mechanisms that log user activity and data mining systems that help predict user behavior on the basis of past behavior can enable direct monitoring of participant behavior, which increases the credibility of electronic marketplaces. Credibility generates trust and a sense of stability among market participants (McKnight et al. 2002).

Second, the ability to use IT competencies effectively requires the knowledge that IT facilitates most strategic actions that, by themselves, do not relate to IT but are required for agile responses to challenges, such as competitive pressures. For example, in a competitive environment, B-to-B electronic marketplaces hire direct sales representatives to market themselves to businesses, as do most firms in the B-to-B industry, to create a point of contact with the potential customer. Many agile marketplaces facilitate the sales process by using web-based referral systems that allow existing buyers (suppliers) to refer the marketplace to external suppliers (buyers). Some marketplaces, such as Autobytel.com, go further by promising to match suppliers to buyers on the basis of not just the inbuilt wholesale car auction mechanism but also through accumulated electronic referrals for both parties in a transaction (Ghose et al. 2007, Business Wire News Release 2013).

Third, managers should recognize the limitations that IT competencies pose when environmental dynamism increases. That is, IT competencies are a source of agility, but they also can create rigidities when the degree of environmental dynamism warrants a continuous reorganization of business functions around integrated IT systems. The piecemeal development of IT infrastructure and skills, without any overarching capability for coordinated action, may foster complacency if independent systems can respond to specific environmental vagaries. But when the dynamism in the whole environment increases, such IT competencies may constrain agile responses. Thus, the strategic allocation of resources for IT competencies should occur only after the firm develops a detailed understanding of the complex ways in which IT interacts with firm-specific and environmental factors that complicate straightforward resource uses. As the short story writer Alice Munro has observed: "The complexity of things—the things within things—just seems to be endless. Nothing is easy, nothing is simple."

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Appendix A. Measures

Entrepreneurial Agility (measured on a 1–7 disagree–agree Likert scale)

1. Our strategy emphasizes building capabilities to foresee a wide range of scenarios.
2. We believe in rapidly taking advantage of opportunities.
3. Our strategic assets can easily be converted to alternate forms.
4. We can easily modify our positioning strategy.

Adaptive Agility (measured on a 1–7 disagree–agree Likert scale)

1. Our organization is pliable, in that we can adjust to abnormal conditions and then bounce right back when conditions come back to normal.
2. Our strategy emphasizes building capabilities to defend against a wide range of scenarios.
3. Our organization can be characterized as resilient.
4. Our organization has the ability to absorb environmental shocks.

IT Infrastructure (measured on a 1–7 disagree–agree Likert scale)

1. We have strong IT planning capabilities.
2. We have extensively invested in building our IT infrastructure.
3. We have a state-of-the-art IT infrastructure.
4. We regularly update our IT assets.

IT Capabilities (measured on a 1–7 disagree–agree Likert scale)

1. We have strong technical IT skills.
2. We have adequate knowledge about IT.
3. Our IT skills are comparable with the best in the industry.
4. We invest heavily in our IT human resources.
5. We have a good understanding of the possible benefits of IT applications.

Firm Performance (measured on a 1–7 unsatisfactory–satisfactory scale)

1. Return on investment relative to objective.
2. Sales relative to objective.
3. Profit relative to goals.
4. Growth relative to objective.
5. General success.

Profiling Variable. Environmental Dynamism (measured on a 1–7 disagree–agree Likert scale)

1. A lot of user firms join and/or leave our marketplace.
2. Our competitors' behaviors exhibit a lot of variability.
3. We are often surprised by our customers' behavior.
4. There are frequent changes in the technical knowledge base of our environment.

Control Variables. Market Orientation (measured on a 1–7 disagree–agree Likert scale)

1. We constantly monitor our level of commitment and orientation to serving customer needs.
2. Our strategy for competitive advantage is based on understanding our customers' needs.
3. We believe that this business exists primarily to serve customers.
4. We measure customer satisfaction systematically and frequently.

5. We freely communicate information about our successful and unsuccessful customer experiences across all business functions.

Market Configuration

Please check the statement that most accurately categorizes the BUSINESS MODEL of your marketplace:

—*Catalogue Aggregator*: The market maker aggregates catalogues (product offerings) of multiple vendors and sells to buyers at a relatively static (fixed) price.

—*Auction*: Multiple buyers bid competitively for products from a single supplier.

—*Reverse Auction*: Buyers post their needs for products or services, and suppliers bid competitively to fulfill the needs.

—*Exchange*: A two-sided marketplace where buyers and sellers interact to exchange information and engage in trade, facilitated through some negotiated dynamic pricing system (such as a bid-and-ask system).

—*Lead Generator*: A seller-driven marketplace, where the market makers derive revenue from advertisements, commission on sales, or fees for generating qualified sales leads for suppliers (sellers).

—*Workflow Marketplace*: Provides project tracking or collaboration services for complex, iterative, multi-party projects (as in construction) and charges fees for its services.

—*Other*: Describe

Vertical/Horizontal Market

A VERTICAL electronic marketplace is one that is specific to an industry (e.g., energy), while a HORIZONTAL marketplace performs the same function (e.g., maintenance) across industries. How would you categorize your marketplace?

— Vertical

— Horizontal

Biased Market

Electronic marketplaces are considered to be either biased or unbiased. In a biased marketplace the market maker (or firms that own the marketplace) transacts in the market as either a buyer or a seller. For example, Covisint is a biased electronic market as a co-owner Ford participates in the market, while ESTEEL is an unbiased electronic marketplace. Please check the statement that most accurately describes the OPERATING status of the ELECTRONIC MARKETPLACE you manage:

—the market maker (or one of the firms that own the marketplace), transacts in the market as a buyer.

—the market maker (or one of the firms that own the marketplace), transacts in the market as a seller.

—the market maker maintains neutrality and does not transact in the electronic market.

Firm Age

How long has the electronic marketplace been in business (years)?

Firm Size

How many total number of employees do you have?

Appendix B. Technical Details on Concomitant Latent Class Regression Analysis (CLCRA)

We provide the details of the CLCRA model used for hypothesis testing in this appendix. Let Y_f be a dependent variable (e.g., firm performance) for firm f , where $f = 1, 2, \dots, F$ are firms. We assume that Y_f

arises from a population that is a finite mixture of $r = 1, 2, \dots, R$ latent regimes in proportions $\theta_1, \theta_2, \dots, \theta_r$ such that (1) $\sum_{r=1}^R (\theta_r) = 1$, and (2) the a priori membership of firms in different regimes is unknown. For the R possible regimes, we specify the relationships between dependent and independent variables as

$$Y_f = \sum_{r=1}^R \theta_r (b_r X'_f + \varepsilon_r), \quad (B1)$$

where X'_f denotes the vector of explanatory variables including the constant term, b_r denotes the vector of regime-specific parameters capturing the effect of the explanatory variables, and ε_r is a regime-specific error term, such that $\varepsilon_r \sim N(0, \sigma_r^2)$. Thus, Y_f has a standard normal density, represented as $f(Y_f | \varphi)$, where $\varphi = [\theta_1, \theta_2, \dots, \theta_R, \phi_1, \phi_2, \dots, \phi_R]'$ and $\phi_r = [b_r, \sigma_r^2]$.

We use a standardized multinomial logit formulation to model θ_r as (Dayton and MacReady 1988):

$$\theta_r = \frac{e^{\lambda_r}}{[1 + \sum_{r=1}^{R-1} e^{\lambda_r}]}, \quad (B2)$$

where λ_r is the regime-specific vector of profiling coefficients. Furthermore, λ_r is parameterized as

$$\lambda_r = \beta_{0r} + \sum_{p=1}^P \beta_{pr} z_{pf}, \quad (B3)$$

where β_{0r} is the regime-specific constant term, and β_{pr} is the profiling coefficient. Because of the standardized multinomial logit formulation, one regime must be treated as the base, and the β_{0r} and β_{pr} coefficients are estimated for the remaining $R - 1$ regimes. In turn, z_{pf} represents the p th profiling variable ($p = 1, 2, \dots, P$). In our context, $P = 1$, because we hypothesize the contingency effects of only one variable, environmental dynamism. By assigning a contingency variable as a profiling variable, we can account explicitly for the effect of the contingency variable in estimating the posterior probabilities of θ_r .

The posterior probability of θ_r offers a way to classify each firm into one of R regimes. For example, if there are three latent regimes of firms, and a firm has posterior probabilities for each regime (0.21 for Regime 1, 0.57 for Regime 2, 0.22 for Regime 3), the algorithm will assign the firm to Regime 2, because it has the highest posterior probability. As per the Bayes rule, the posterior probability that firm f belongs to regime r is given by

$$P[f \in r | Y_f] = \frac{\theta_r L_{f|r}}{\sum_{r=1}^R \theta_r L_{f|r}}, \quad (B4)$$

where $L_{f|r}$ is the likelihood that firm f belongs to regime r .

The likelihood is specified by the standard normal density $f(Y_f | \varphi)$, where φ is a vector of all parameters to be estimated. Thus, the likelihood function that needs to be maximized for parameter estimation is:

$$L(\varphi, Y) = \prod_{f=1}^F f(Y_f | \varphi). \quad (B5)$$

The natural logarithm of this equation can be maximized with an expectation-maximization (EM) algorithm

(Dempster and Laird 1977), which provides the parameter estimates for an R -regime solution. For example, if we prespecify the number of regimes as two, maximizing the natural logarithm of Equation (B5) provides a two-regime solution for φ . A prespecification of three regimes provides a three-regime solution for φ , and so on. Because we do not know a priori the number of latent regimes that will fit our model, the obvious question becomes how to determine R . We use an information criterion to assess the number of regimes that provides the best possible model fit. Consistent with the Monte Carlo simulation study of Andrews and Currim (2003), we use a modified Akaike information criterion (MAIC) to determine the number of latent regimes. Specifically, the MAIC is (where P denotes the number of parameters to be estimated and L is the log-likelihood function):

$$MAIC = -2 * \ln(L) + 3 * P. \quad (B6)$$

The number of regimes that provides the lowest value of MAIC is selected as the model with the best possible fit. Therefore, we start with the homogenous solution ($R = 1$, which is a standard regression model), then estimate the two-regime solution (i.e., $R = 2$). If model fit improves (i.e., MAIC decreases), we estimate the solution with one more regime (i.e., $R = 3$). We continue to add regimes until there is no improvement in model fit in terms of the information criterion.

To validate whether MAIC leads to the appropriate selection of the number of regimes, we can examine an entropy measure of separation (ES). If the latent class model with R regimes fits the data well, each firm likely has a high posterior probability of belonging to one regime but low posterior probabilities of belonging to other $R - 1$ regimes. In this case, ES is a measure of the certainty of the separation in posterior probabilities (Dias and Vermunt 2007), ranging in $[0, 1]$, such that values closer to 1 indicate good separation. The ES values tend to be lower (close to 0) when the probability of belonging to a regime is close to $1/R$, where R is the number of regimes. We calculate ES as follows:

$$ES = 1 - \frac{\sum_{f=1}^F \sum_{r=1}^R -p_{f \in r | Y_f} \ln(p_{f \in r | Y_f})}{F \ln(R)}, \quad (B7)$$

where $p_{f \in r | Y_f}$ is the posterior probability of firm f belonging to regime r , which we calculate using the Bayes rule specified in Equation (4).

With any given sample size, there is always a possibility in CLCRA that we can identify regimes of varying sizes (e.g., Grewal et al. 2008, Srinivasan 2006); the likelihood of identifying regimes with few firms increases as sample size decreases (we have a relatively small sample of 109 firms, which is approximately 25% of our population). However, we are not estimating independent models across regimes; instead, we estimate one model with multiple regimes, so we can use the whole sample to estimate coefficients for all regimes, large and small. Because CLCRA is well established in statistics (e.g., Dayton and MacReady 1988), psychometrics (e.g., Wedel and DeSarbo 1998), marketing (e.g., Srinivasan 2006), and management science (e.g., Grewal et al. 2006) literature, it already has been shown to provide unbiased, consistent, and efficient parameter estimates for regimes with many and few observations (or regimes).

Appendix C. Latent Class Confirmatory Factor Analysis to Establish Measurement Invariance

Typically, the measurement model is represented as:

$$x = \nu + \Lambda_x \xi + \delta, \quad (C1)$$

where x is a $(q \times 1)$ vector of observable indicator variables, ξ is the $(n \times 1)$ vector of latent constructs, Λ_x is the $(q \times n)$ matrix of factor loadings, ν is the $(p \times 1)$ vector of measurement intercepts, and δ is $(p \times 1)$ vector of measurement errors. In the preceding equation, ξ is a vector of latent constructs both Y and X variables in Equation (B1). For c latent classes, a latent class-specific measurement model can be written as (e.g., Yung 1997):

$$x | c = \nu^c + \Lambda_x^c \xi^c + \delta^c, \quad (C2)$$

where $E(\xi^c) = \tau^c$, $E[(\xi^c - \tau^c)(\xi^c - \tau^c)'] = \Phi^c$, $E(\delta^c \delta^{c'}) = \Theta_\delta^c$, and we assume that $E(\delta^c) = 0$. Thus, conditional on latent class membership, the conditional mean vector (μ^c) is:

$$\mu^c = \nu^c + \Lambda_x^c \xi^c. \quad (C3)$$

The corresponding covariance matrix Σ^c ($q \times q$) is given by:

$$\Sigma^c = \Lambda_x^c \Phi^c \Lambda_x^{c'} + \Theta_\delta^c. \quad (C4)$$

We assume $\Delta | c$, the joint vector of observable indicators that is conditional on latent class c , reveals a multivariate normal distribution, which provides the unconditional distribution of Δ as a finite mixture of these conditionals; that is,

$$\Delta \sim \sum_{c=1}^C \kappa^c f^c(\Delta | \mu^c \Sigma^c), \quad (C5)$$

where $\kappa = (\kappa^1, \kappa^2, \dots, \kappa^C)'$ is a vector of C mixing proportions, such that $\kappa^c > 0 \forall c = 1, 2, \dots, C$, and $\sum_{c=1}^C \kappa^c = 1$. Thus, the likelihood function for c latent classes and sample size of N can be written as:

$$\prod_{i=1}^N \left[\sum_{c=1}^C \kappa^c (2\pi)^{-q/2} |\Sigma^c|^{-1/2} e^{-1/2(\Delta_i - \mu^c)'(\Sigma^c)^{-1}(\Delta_i - \mu^c)} \right]. \quad (C6)$$

Again, similar to CLCRA, we use the EM algorithm to maximize the log likelihood, calculate posterior latent class membership using the Bayes rule, and evaluate the entropy of separation. We use the Bayesian information criterion (BIC) to determine the optimal number of latent regimes (Yung 1997). Measurement invariance exists if a single latent class solution is optimal.

References

- Achrol RS, Stern LW (1988) Environmental determinants of decision-making uncertainty in marketing channels. *J. Marketing Res.* 25(1):36–50.
- Agarwal R, Gort M (2002) Firm and product life cycles and firm survival. *Amer. Econom. Rev.* 92(2):184–190.
- Albrecht CC, Dean DL, Hansen JV (2005) Marketplace and technology standards for B2B e-commerce: Progress, challenges, and the state of the art. *Inform. Management* 42(6):865–875.
- Anderson JC, Gerbing DW (1988) Structural equation modeling in practice: A review and recommended two-step approach. *Psych. Bull.* 103(3):411–423.

- Andrews RL, Currim IS (2003) Retention of latent segments in regression-based marketing models. *Internat. J. Res. Marketing* 20(4):315–321.
- Aral S, Weill P (2007) IT assets, organizational capabilities, and firm performance: How resource allocations and organizational differences explain performance variation. *Organ. Sci.* 18(5):763–780.
- Armstrong JS, Overton TS (1977) Estimating nonresponse bias in mail surveys. *J. Marketing Res.* 14(3):396–402.
- Bagozzi RP, Yi Y (1988) On the evaluation of structural equation models. *J. Acad. Marketing Sci.* 16(Spring):74–94.
- Banker RD, Bardhan IR, Chang H, Lin S (2006) Plant information systems, manufacturing capabilities and plant performance. *MIS Quart.* 30(2):315–337.
- Bapna R, Barua A, Mani D, Mehra A (2010) Coordination and cooperation, coordination, and governance in multisourcing: An agenda for analytical and empirical research. *Inform. Systems Res.* 21(December):785–795.
- Barrett H, Weinstein A (1998) The effect of market orientation and organizational flexibility on corporate entrepreneurship. *Entrepreneurship: Theory and Practice* 23(1):57–70.
- Bartlett MS (1937) The statistical conception of mental factors. *British J. Psych.* 28:97–104.
- Barua A, Mukhopadhyay T (2000) Information technology and business performance. Zmud RW, ed. *Framing the Domains of IT Management: Projecting the Future Through the Past*. (Pinnaflex Press, Cincinnati).
- Barua A, Kriebel CH, Mukhopadhyay T (1995) Information technologies and business value: An analytic and empirical investigation. *Inform. Systems Res.* 6(1):3–23.
- Bharadwaj AS (2000) A resource-based perspective on information technology capability and firm performance: An empirical investigation. *MIS Quart.* 24(1):169–196.
- Bharadwaj AS, Bharadwaj SG, Konsynki BR (1999) Information technology effects on firm performance as measured by Tobin's q . *Management Sci.* 45(7):1008–1024.
- Brynjolfsson E, Hitt L (1996) Paradox lost? Firm-level evidence on the returns to information systems spending. *Management Sci.* 42(4):541–558.
- Business Wire News Release (2013) Press release (April 10, 2013), <http://finance.yahoo.com/news/patti-scipione-m-lady-nissan-123000031.html>.
- Celeux G, Soromenho G (1996) An entropy criterion for assessing the number of clusters in a mixture model. *J. Classification* 13(2):195–212.
- Chandy RK, Tellis GJ (2000) The incumbent's curse? Incumbency, size, and radical product innovation. *J. Marketing* 64(3):1–17.
- Chi L, Ravichandran T, Andreovski G (2010) Digital platforms, network structure and competitive actions. *Inform. Systems Res.* 21(3):543–570.
- Chiasson MW, Davidson E (2005) Taking industry seriously in information systems research. *MIS Quart.* 29(4):591–605.
- D'Aveni RA, Battista D, Smith KG (2010) The age of temporary advantage. *Strategic Management J.* 31(13):1371–1385.
- Dayton MC, MacReady GB (1988) Concomitant variables latent class models. *J. Amer. Statist. Assoc.* 83(401):173–178.
- Dempster AP, Laird MN (1977) Maximum likelihood estimation from incomplete data via the E-M algorithm. *J. Royal Statist. Soc. Series B* (39):1–38.
- Deshpandé R, Farley JU (1998) Measuring market orientation: Generalization and synthesis. *J. Market-Focused Management* 2(3):213–232.
- Dias JG, Vermunt JK (2007) Latent class modeling of website users' search patterns: Implications for online market segmentation. *J. Retailing and Consumer Services* 14(6):359–368.

- Fornell C, Larcker DF (1981) Structural equation models with unobservable variables and measurement error: Algebra and statistics. *J. Marketing Res.* 18(3):382–388.
- Garen J (1984) The returns to schooling: A selectivity bias approach with a continuous choice variable. *Econometrica* 52(5):1199–1218.
- Garg VK, Walters BA, Priem RL (2003) Chief executive scanning emphases, environmental dynamism, and manufacturing firm performance. *Strategic Management J.* 24(8):725–744.
- Garicano L, Kaplan SN (2001) The effects of business-to-business e-commerce on transaction costs. *J. Indust. Econom.* 49(December):463–485.
- Ghose A, Mukhopadhyay T, Rajan U (2007) The impact of Internet referral services on a supply chain. *Inform. Sys. Res.* 18(3):300–319.
- Goldman SL, Nagel RN, Preiss K (1995) *Agile Competitors and Virtual Organizations: Strategies for Enriching the Customer* (Van Nostrand Reinhold, New York).
- Grewal R, Tansuhaj P (2001) Building organizational capabilities for managing economic crises: The role of market orientation and strategic flexibility. *J. Marketing* 65(April):67–80.
- Grewal R, Lilien G, Mallapragada G (2006) Location, location, location: How network embeddedness affects project success in open source systems. *Management Sci.* 52(7):1043–1056.
- Grewal R, Chakravarty A, Ding M, Liechty J (2008) Counting chickens before the eggs hatch: Associating new product development portfolios with shareholder expectations in the pharmaceutical sector. *Internat. J. Res. Marketing* 25(4):261–272.
- Grewal R, Chandrashekar M, Johnson JL, Mallapragada G (2013) Moderating role of the environment for the effect of market orientation on firm outcomes: Accounting for unobserved heterogeneity. *J. Acad. Marketing Sci.* 41(2):206–233.
- Grimm CM, Lee H, Smith KG (2006) *Strategy as Action: Competitive Dynamics and Competitive Advantage* (Oxford University Press, New York).
- Hitt MA, Keats BW, DeMarie SM (1998) Navigating in the new competitive landscape: Building strategic flexibility and competitive advantage in the 21st century. *Acad. Management Executive* 12(4):22–42.
- Hough JR, White MA (2003) Environmental dynamism and strategic decision-making rationality: An examination at the decision level. *Strategic Management J.* 24(5):481–489.
- Hurley RF, Hult TM (1998) Innovation, market orientation, and organizational learning: An integration and empirical examination. *J. Marketing* 62(3):42–54.
- Jaworski BJ, Kohli AK (1993) Market orientation: Antecedents and consequences. *J. Marketing* 57(3):53–70.
- Joshi K (1991) A model of users' perspective on change: The case of information systems technology implementation. *MIS Quart.* 15(2):229–242.
- Kambil A, van Heck E (2002) *Making Markets: How Firms Can Design and Profit from Online Auctions and Exchanges* (Harvard Business School Press, Boston).
- Kaplan SN, Sawhney M (2000) E-hubs: The new B2B marketplace. *Harvard Bus. Rev.* 78(May/June):97–103.
- Kirca AH, Jayachandran S, Bearden WO (2005) Market orientation: A meta-analytic review and assessment of its antecedents and impact on performance. *J. Marketing* 69(2):24–41.
- Koch H (2010) Developing dynamic capabilities in electronic marketplaces: A cross-case study. *J. Strategic Inform. Systems* 19(1):28–38.
- Kohli R, Grover V (2008) Business value of IT: An essay on expanding research directions to keep up with the times. *J. Assoc. Inform. Systems* 9(1):23–39.
- Lastovicka JL, Thamodaran K (1991) Common factor score estimates in multiple regression problems. *J. Marketing Res.* 28(1):105–112.
- Lee D, Sambamurthy V, Lim K, Wei KK (2003) The moderating effects of environmental dynamism on the links between IT management and agility: A moderated mediation analysis, Working paper, University of Massachusetts, Boston.
- Leisch F (2004) FlexMix: A general framework for finite mixture models and latent class regression in R. *J. Statist. Software* 11(8):1–18.
- Lindell MK, Whitney DJ (2001) Accounting for common method variance in cross-sectional research designs. *J. Appl. Psych.* 86(1):114–121.
- Lu Y, Ramamurthy K (2011) Understanding the link between information technology capability and organizational agility: An empirical examination. *MIS Quart.* 35(4):931–954.
- Malhotra Y (2003) Why knowledge management systems fail? Enablers and constraints of knowledge management in human enterprises. Koenig MED, Srikantaiah TK, eds. *Knowledge Management Lessons Learned: What Works and What Doesn't* (Medford, New Jersey).
- Malone TW, Yates J, Benjamin RI (1989) The logic of electronic markets. *Harvard Bus. Rev.* 67(3):166–172.
- Mata F, Fuerst WL, Barney JB (1995) Information technology and sustained competitive advantage: A resource-based analysis. *MIS Quart.* 19(4):487–505.
- McKnight DH, Choudhury V, Kacmar C (2002) Developing and validating trust measures for e-commerce: An integrative typology. *Inform. Systems Res.* 13(3):334–359.
- Meredith W (1993) Measurement invariance, factor analysis and factorial invariance. *Psychometrika* 58(4):525–543.
- Mishina Y, Pollock TG, Porac JF (2004) Are more resources always better for growth? Resource stickiness in market and product expansion. *Strategic Management J.* 25(12):1179–1197.
- Muthén B (1989) Latent variable modeling in heterogeneous populations. *Psychometrika* 54(4):557–585.
- Muthén B (2001) *Latent Variable Mixture Modeling* (Mahaw, NJ).
- Muthén B, Shedden K (1999) Finite mixture modeling with mixture outcomes using the EM algorithm. *Biometrics* 55(2):463–469.
- Muthén L, Muthén B (2007) *Mplus User's Guide* (Muthé and Muthé, Los Angeles).
- Overby E, Bharadwaj A, Sambamurthy V (2006) Enterprise agility and the enabling role of information technology. *Eur. J. Inform. Systems* 15:120–131.
- Pavlou PA, El Sawy OA (2006) From IT leveraging competence to competitive advantage in turbulent environments: The case of new product development. *Inform. Systems Res.* 17(3):198–227.
- Pavlou PA, El Sawy OA (2010) The “third hand”: IT-enabled competitive advantage in turbulence through improvisational capabilities. *Inform. Systems Res.* 21(3):443–471.
- Pavlou PA, Gefen D (2004) Building effective online marketplaces with institution-based trust. *Inform. Systems Res.* 15(1):37–59.
- Peteraf MA (1993) The cornerstones of competitive advantage: A resource-based view. *Strategic Management J.* 14(3):179–191.
- Piccoli G, Ives B (2005) IT-dependent strategic initiatives and sustained competitive advantage: A review and synthesis of literature. *MIS Quart.* 29(4):747–776.
- Podsakoff PM, Organ DW (1986) Self-reports in organizational research: Problems and prospects. *J. Management* 12(4):531–544.
- Rai A, Patnayakuni R, Patnayakuni N (2006) Firm performance impacts of digitally enabled supply chain integration capabilities. *MIS Quart.* 30(2):225–246.
- Rindova VP, Kotha S (2001) Continuous “morphing”: Competing through dynamic capabilities, form, and function. *Acad. Management J.* 44(6):1263–1280.
- Ross JW, Beath CM, Goodhue DL (1996) Develop long-term competitiveness through IT assets. *Sloan Management Rev.* 38(1):31–42.

- Sambamurthy V, Zmud RW (1999) Arrangements for information technology governance: A theory of multiple contingencies. *MIS Quart.* 23(2):261–290.
- Sambamurthy V, Bharadwaj AS, Grover V (2003) Shaping agility through digital options: Reconceptualizing the role of information technology in contemporary firms. *MIS Quart.* 27(2):237–263.
- Sanchez R, Mahoney JT (2003) Modularity, flexibility, and knowledge management in product organization design. Raghu GAK, Kumaraswamy A, Langlois RN, eds. *Managing the Modular Age: Architectures, Networks and Organization Design* (Basil Blackwell Publishers, New York), 362–380.
- Sarkar MB, Echambadi R, Harrison JS (2001) Alliance entrepreneurship and firm market performance. *Strategic Management J.* 22(6–7):701–711.
- Sethi R, Pant S, Sethi A (2003) Web-based product development systems integration and new product outcomes: A conceptual framework. *J. Product Innovation Management* 20(1):37–56.
- Sheffi Y, Rice JBJ (2005) Building the resilient enterprise. *MIT Sloan Management Rev.* 47(1):41–48.
- Sirmon DG, Hitt MA, Ireland DR (2007) Managing firm resources in dynamic environments to create value: Looking inside the black box. *Acad. Management J.* 32(1):273–292.
- Skrondal A, Laake P (2001) Regression among factor scores. *Psychometrika* 66(4):563–576.
- Smith MD, Bailey J, Brynjolfsson E (2000) Understanding digital markets: Review and assessment. Brynjolfsson E, Kahin B, eds. *Understanding the Digital Economy: Data, Tools and Research* (MIT Press, Cambridge, MA), 276–288.
- Srinivasan R (2006) Dual distribution and intangible firm value: Franchising in restaurant chains. *J. Marketing* 70(3):120–135.
- Tallon PP, Pinsonneault A (2011) Competing perspectives on the link between strategic information technology alignment and organizational agility: Insights from a mediation model. *MIS Quart.* 35(2):463–486.
- Teece DJ (2007) Explicating dynamic capabilities: The nature and microfoundations of sustainable enterprise performance. *Strategic Management J.* 28(13):1319–1350.
- Thatcher JB, Perrewe PL (2002) An empirical examination of individual traits as antecedents to computer anxiety and computer self-efficacy. *MIS Quart.* 26(4):381–396.
- Thomson GH (1934) The meaning of i in the estimate of g . *British J. Psych.* 25:92–99.
- Titterton D, Smith A, Makov U (1985) *Statistical Analysis of Finite Mixture Distributions*. (John Wiley & Sons, Chichester, England).
- United States Census Bureau (2010) E-Stats (May 10, 2010), <http://www.census.gov/econ/estats/2010/2010reportfinal.pdf>.
- Wedel M, DeSarbo WS (1998) Mixtures of (constrained) ultrametric trees. *Marketing Lett.* 63(4):17–25.
- Wheeler B (2002) NeBIC: A dynamic capabilities theory for assessing net-enablement. *Inform. Systems Res.* 13(2):125–146.
- Winter SG (2003) Understanding dynamic capabilities. *Strategic Management J.* 24(10):991–995.
- Wixom BH, Watson HJ (2001) An empirical investigation of the factors affecting data warehousing success. *MIS Quart.* 25(1):17–41.
- Worren N, Moore K, Cardona P (2002) Modularity, strategic flexibility, and firm performance: A study of the home appliance industry. *Strategic Management J.* 23(12):1123–1140.
- Yung YF (1997) Finite mixtures in confirmatory factor-analysis models. *Psychometrika* 62(3):297–330.